Scalability & Future Plan

Date	12 July 2026
Team ID	SB-HDI-07
Project Name	Human Development Index (HDI) Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the Human Development Index (HDI) Prediction System can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Model is trained on a static, historical UNDP HDI dataset with no live data refresh	Predictions can become outdated as new socio-economic indicators are published each year	High
2	Application runs as a single Flask/Streamlit instance with no load balancing	Cannot serve a large number of concurrent users (e.g., during classroom or organization-wide demos) without slowdowns or crashes	High
3	No caching layer; every prediction re-runs full feature preprocessing and model inference	Increases response latency and redundant compute cost for repeated or similar queries	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	**User Load**	Single local server handling limited, sequential prediction requests	Containerize the app with Docker and deploy on a cloud platform (AWS/GCP/Azure) behind a load balancer with auto-scaling instances to handle traffic spikes
2	**Data Storage**	HDI indicators stored in a static local CSV file	Migrate to a managed cloud database (PostgreSQL/MongoDB) with a scheduled ETL job to pull the latest indicators from the UNDP HDI API/reports
3	**Performance**	Synchronous, uncached inference on every request using the raw trained model file	Introduce a caching layer (Redis) for repeated country/scenario queries, export the model to an optimized format (ONNX), and support asynchronous/batch prediction requests

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
4	**Security**	Open endpoint with no authentication or input validation	Add API authentication (JWT/OAuth2), enforce HTTPS, validate and sanitize all user-provided indicator inputs, and apply rate limiting to prevent abuse

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate live UNDP/World Bank data feeds via API so indicators (life expectancy, GNI, schooling years) refresh automatically	Q3 2026	Improves prediction accuracy and keeps HDI classifications current without manual dataset updates
Phase 3	Build a multi-country comparative dashboard with interactive charts and trend visualizations	Q4 2026	Enables policymakers and researchers to benchmark countries side by side and spot development gaps faster
Phase 4	Extend model to sub-national/regional HDI prediction and release a mobile app version	Q1 2027	Widens accessibility, supports finer-grained policy targeting within large countries